Do Now · retrieval

Database and SQL retrieval challenge

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is a primary key?

Q2

What does INNER JOIN return?

Q3

What is a foreign key?

Q4

A 3×2 zero-indexed array: specify row index range, column range and cell count.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Understand — Use a primary key for each Student row and a matching foreign key in Score.

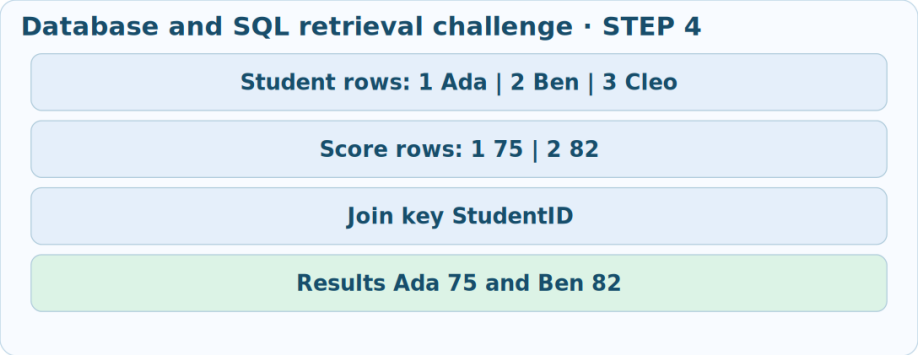
Core idea 2

Trace — INNER JOIN produces one result for each matching StudentID; Cleo has no matching Score and is absent.

Core idea 3

Evaluate — A WHERE clause can filter matched records; forgetting the join condition can produce unwanted combinations.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

primary key — Unique row identifier.

foreign key — Related table key reference.

INNER JOIN — Returns matching joined rows.

WHERE — Filters query results.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

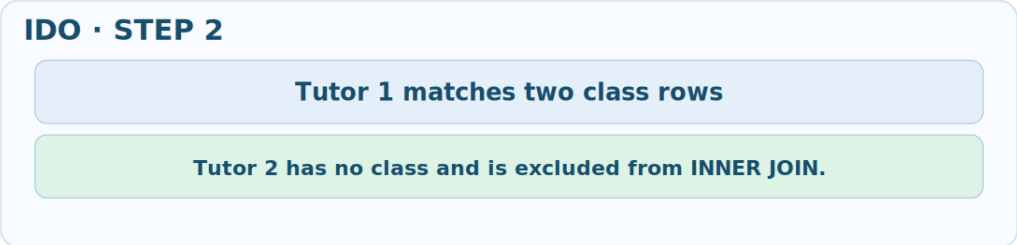
Problem

Match Tutor and Class records on TutorID.

Follow the worked method

1. Predict first — Tutor 1 matches two class rows
2. Model and check — Tutor 2 has no class and is excluded from INNER JOIN.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first
IDs 4 and 6 match

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Define primary key, foreign key, INNER JOIN and WHERE. [4]

Question 2

Write an INNER JOIN for Student and Score above selecting Name and Mark; give output. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Explain why omitting a join condition may produce misleading results. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

INNER JOIN automatically includes rows without a key match.

Use key vocabulary from memory

1. Explain primary key in your own words.

2. Explain foreign key in your own words.

3. Explain INNER JOIN in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.